VIRTUALIZATION & VIRTUAL MACHINES

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VMWARE ECOSYSTEM

VIRTUALIZATION

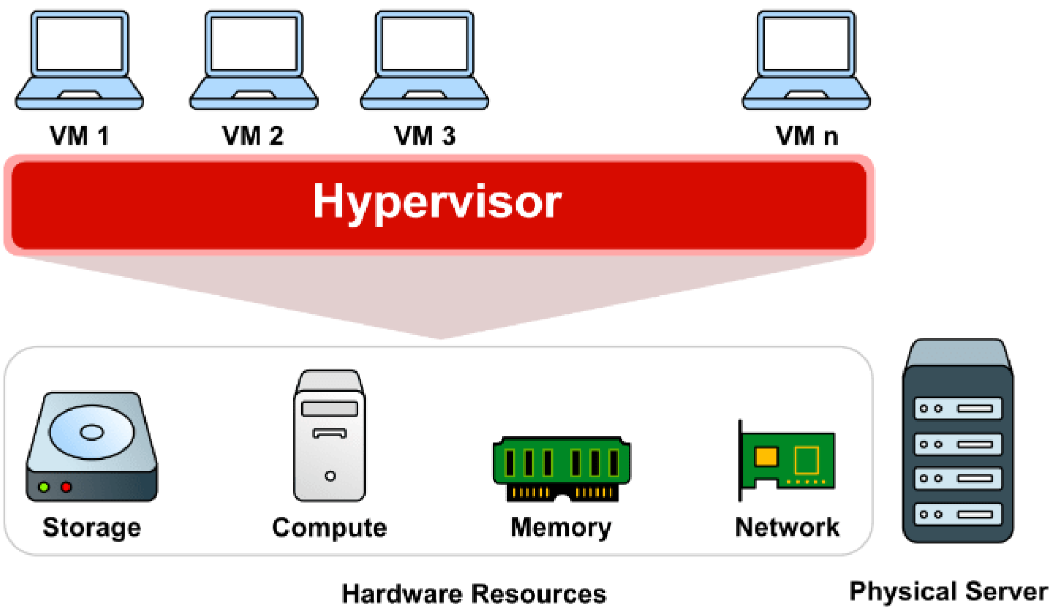
VIRTUAL MACHINES (VM)

Virtualization is the technology that allows the creation of a Virtual Version of a physical resource, such as a server, storage device, or a network component. The abstraction of hardware and software resources enables multiple computing environments to run on a single physical machine, improving efficiency and flexibility.

A Virtual Machine (VM) is software that creates a computer environment within another computer. Each VM has its own operating system and operates independently of other VMs, even if they are on the same physical computer.

Virtual Machines operate through software called Hypervisor, which manages hardware resources (CPU, Memory, Storage Devices) and allocates them to VMs. The hypervisor allows multiple operating systems to run on the same physical computer.

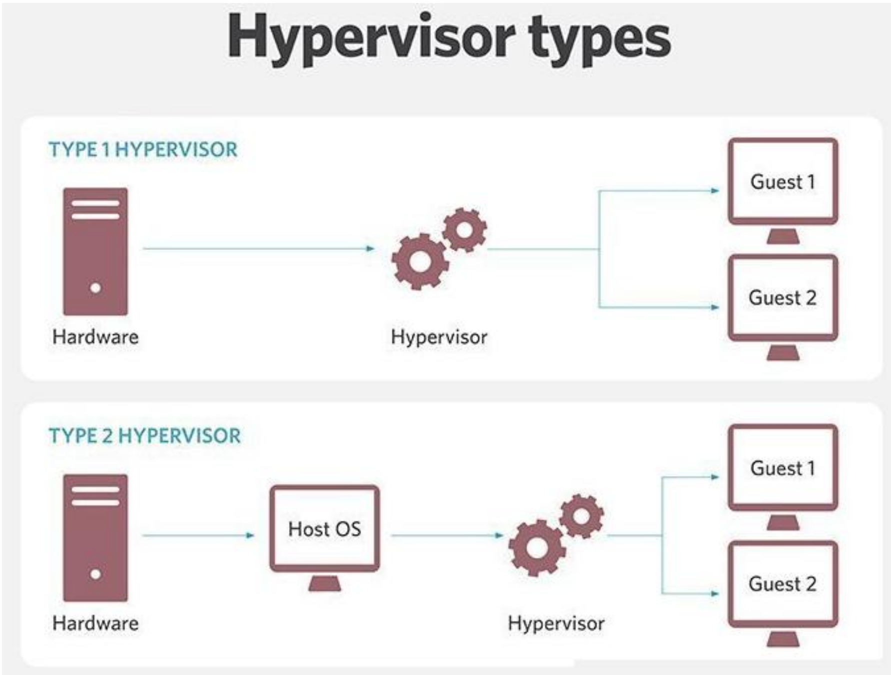
The concept of virtual machine started back in the 60's with IBM, which developed the technology to allow multiple operating systems to run on a mainframe computer. Since then, the technology has evolved significantly and is widely used in various fields.



A Physical Server with Multiple VMs

There are two types of Hypervisors:

A. ***Type 1*** *(Bare Metal)*: These run on the computer's hardware and offer high performance and security. Examples include *VMware ESXi* and *Microsoft Hyper-V*.
B. ***Type 2*** *(Hosted)*: These run on top of an Operating System and are easier to use and install. Examples include *VMware Workstation* and *Oracle VirtualBox*.



Bare Metal Hypervisor vs Hosted Hypervisor

In nowadays the use of VMs is mandatory in many fields, such as:

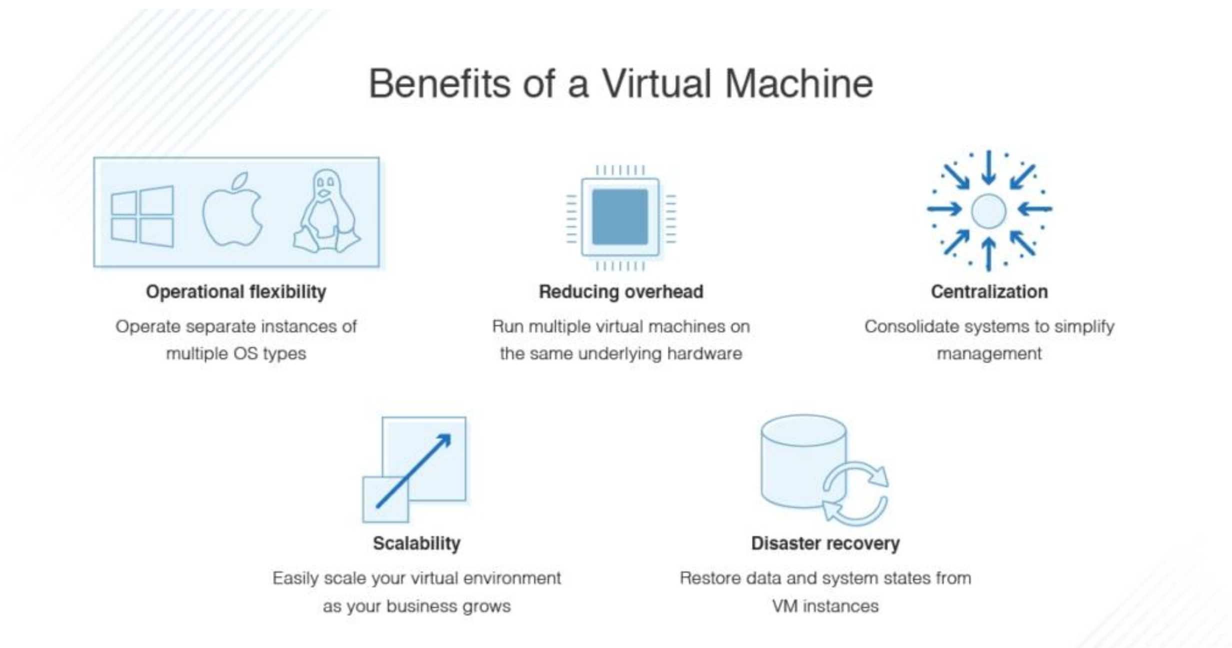
- Software Development and Testing: Developers can create and test applications on different operating systems without needing multiple physical computers.
- Cloud Infrastructure: Cloud service providers use VMs to offer flexible and scalable services to their customers.
- Educations and Learning: Educators can create virtual environments to teach various operating systems and applications.
- Security and Data Recovery: VMs can be used to isolate and analyze malware without affecting the main system.

As an example of use cases:

- Google: Uses VMs to manage the resources of its data centers.
- Netflix: Uses VMs to provide streaming services to the users.

Virtual Machines offer many advantages:

- Cost and Efficiency: They reduce the need for multiple physical computers, saving space and maintenance costs.
- Flexibility and Portability: VMs can be easily moved from one hypervisor to another, allowing flexibility in resource management.
- Isolation and Security: Each VM is isolated from others, reducing the risk of malware spreading.



The technology of VMs continues to evolve rapidly. In the future we are going to find it in many more installations.

- Integration with Artificial Intelligence: VMs will be used for training and developing AI models.
- Advanced Cloud Services: Cloud providers will offer more specialized and customized services through VMs.
- Enhanced Security: New technologies will be integrated to improve the security of VMs and protect data.

VMWARE (VMWARE)

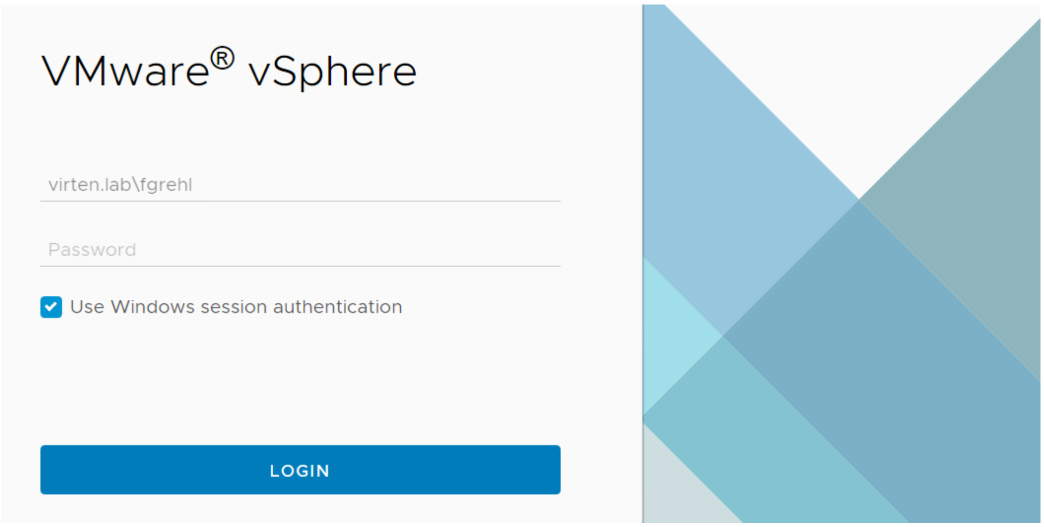
VMware is a leading technology company specializing in virtualization and cloud solutions. It offers many products but the most important for us are, VMware Workstation, VMware vSphere, VMware ESXi and vCenter.

VMWARE VSPHERE

Is a flagship virtualization platform, designed to transform data centers into combined computing infrastructures that include CPU, storage, and networking resources. vSphere is a suite of products, just like Microsoft Office. It manages these infrastructures as a unified operating environment, providing tools to administer and optimize data centers.

Most important vSphere components are:

1. VMware ESXi: The hypervisor that runs directly on physical servers, allowing the creation and management of multiple VMs on a single host.
2. vCenter Server: The centralized management platform for vSphere environments, enabling the management of multiple ESXi hosts and VMs from single interface.



vSphere Login Page

The vSphere is used in many cases such as:

- Data Center Consolidation: Reduces hardware costs and improves resource utilization by consolidating multiple physical servers into fewer, more powerful hosts running multiple VMs.
- Disaster Recovery: Enhances business continuity with features like High Availability, which ensures minimal downtime by automatically restart VMs on other hosts in the event of a host failure and vMotion, which allows live migration of VMs from one host to another without downtime.
- Development and Testing: Provides isolated environments for software development and testing, allowing developers to quickly spin up and tear down VMs.
- Cloud Integration: Integrates seamlessly with public and private cloud environments, enabling hybrid cloud strategies.

What are the advantages of using vSphere in a company or organization:

- Efficiency: Optimizes resource usage and reduces operational costs.
- Scalability: Easily scales to meet growing business demands.
- Security: Enhances security with features like secure boot, encryption, and role-based access control.

VMWARE ESXI (VMWARE ESXI INSTALLATION & CONFIGURATION)

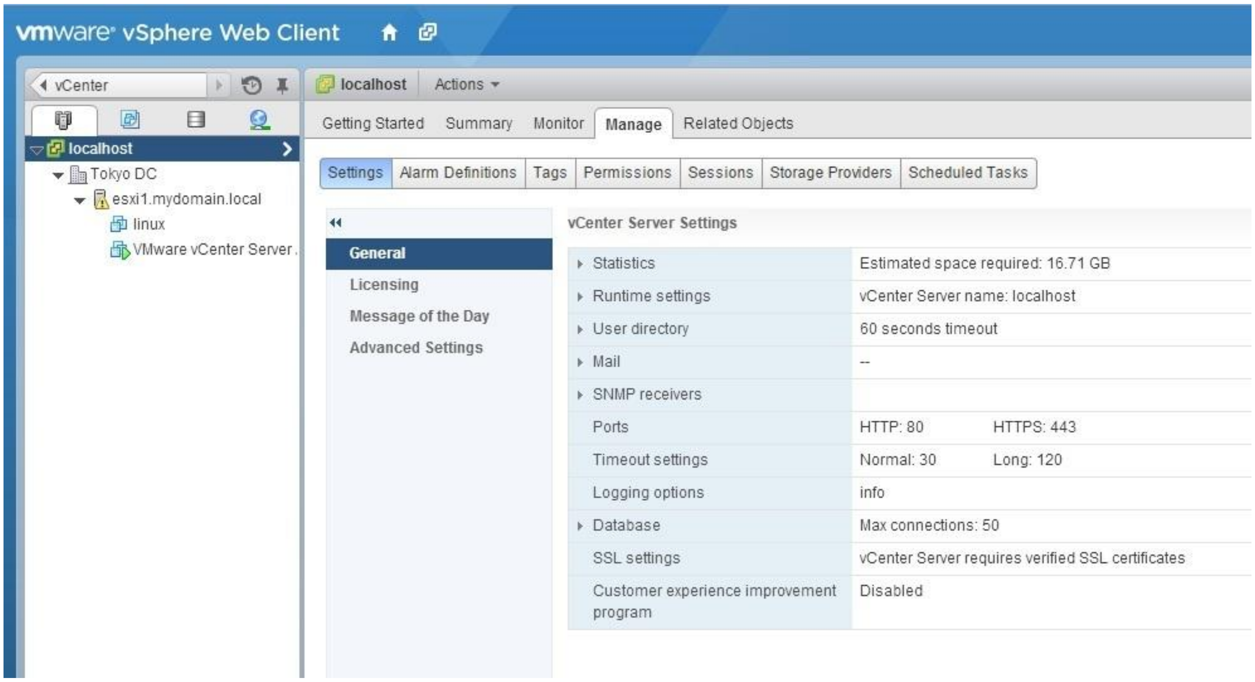
VMware ESXi is the core of the vSphere product suite. It is a *Bare-metal Hypervisor* that allows the creation and management of virtual machines. It is widely used in data center for combining and managing computing, storage, and networking resources.

The VMware ESXi provides, robust security features such as Secure Boot and TPM (Trusted Platform Module). It supports high performance with capabilities like vMotion (Allows the migration of VMs from one ESXi host to another without downtime) and DRS (Distributed Resource Scheduler). DRS automatically allocates resources among VMs for optimal performance. Finally it has easy management through vCenter Server, which allows centralized management of multiple ESXi hosts.

VMware ESXi 5.0

VCENTER

The VMware vCenter is a product under the VMware vSphere suite. Is a central management tool used to control and monitor VMs created in the VMware ESXi hypervisor. The vCenter provides a centralized management of VMs and ESXi hosts all from a single interface.



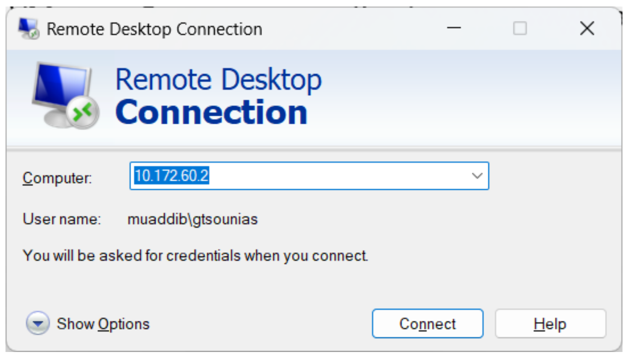
VMware vCenter

VCENTER

In order to be able to access the VMware vCenter, we first have to create each user in Active Directory in Domain Controller.

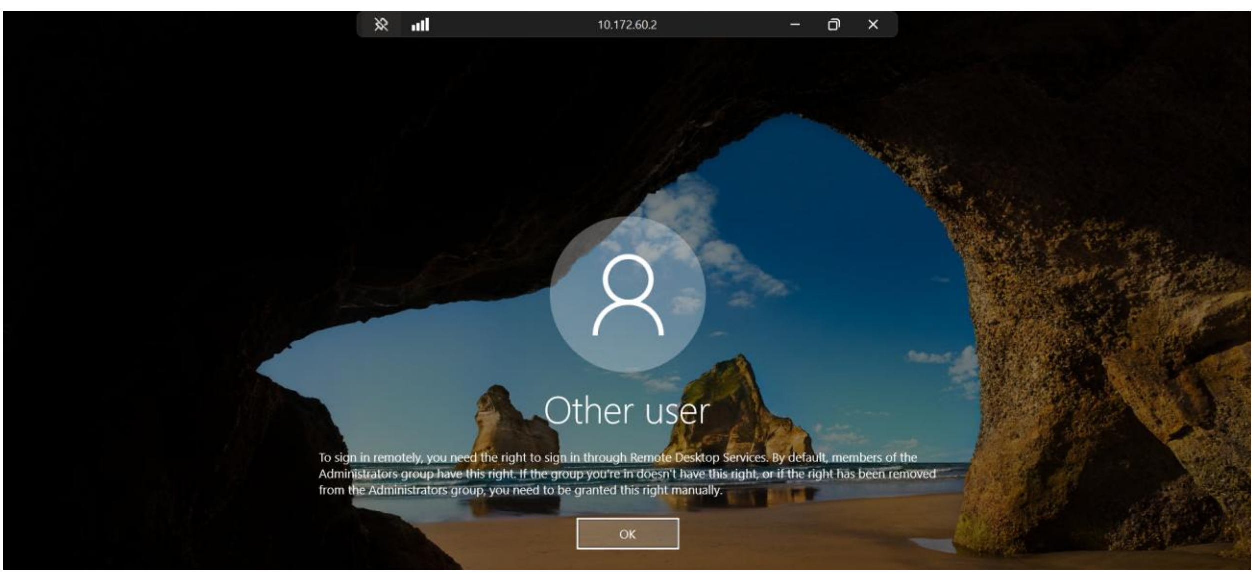
Attention: Only the ***Admins*** have access on DC server for configuration

- RDP to Domain Controller: 10.172.60.2

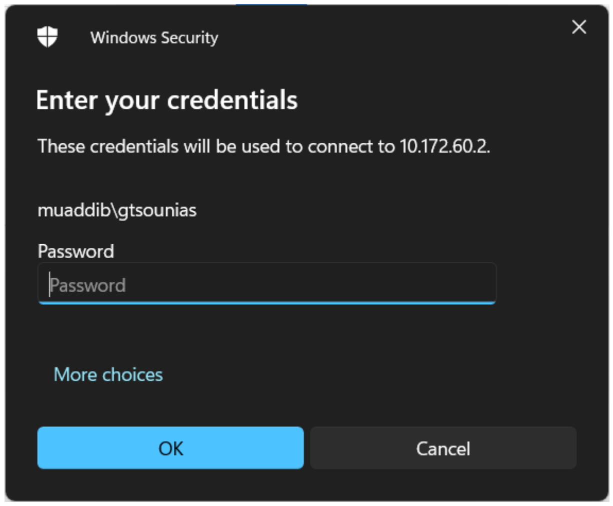


Click Connect

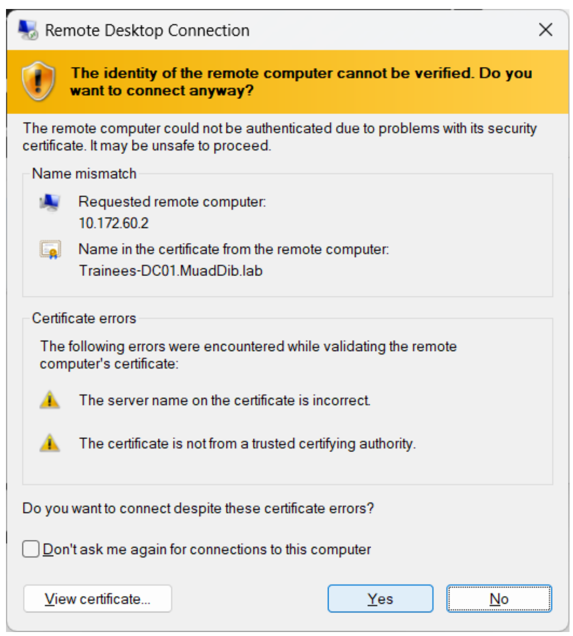
Error Page:



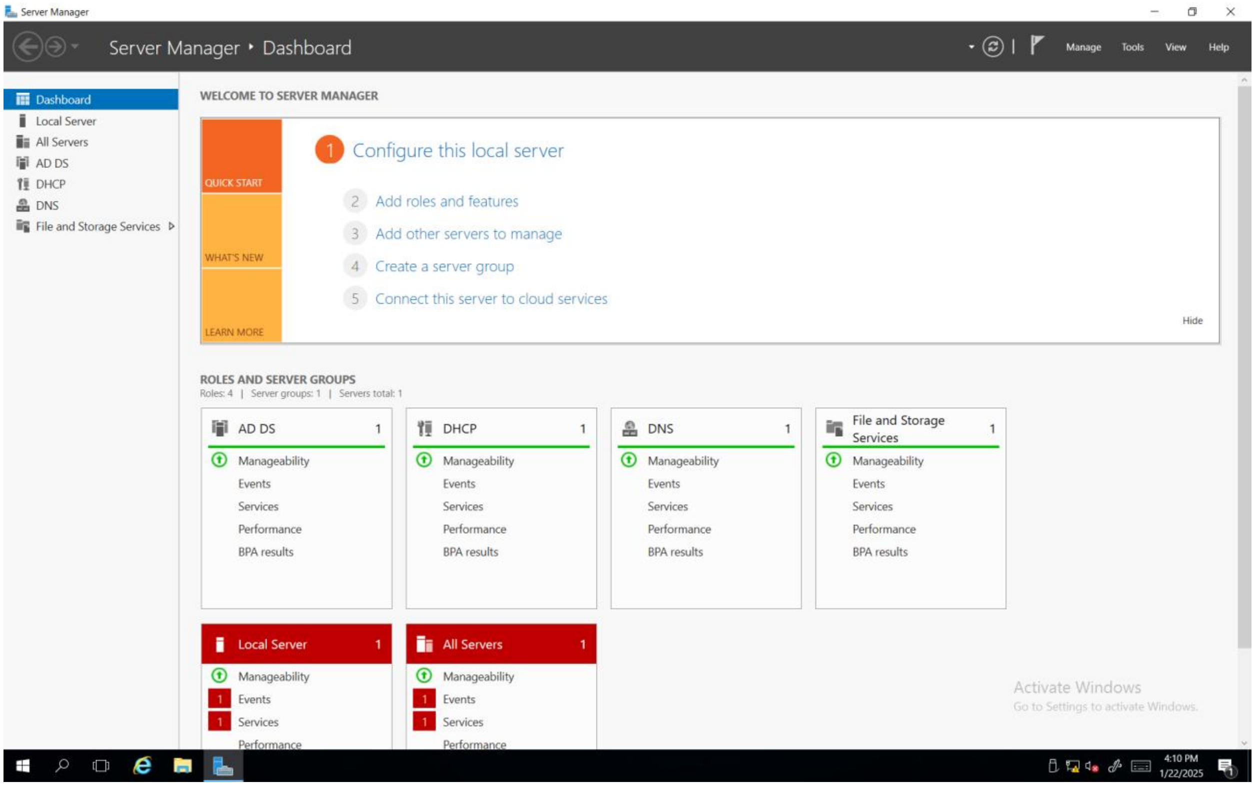
This is the screen that a NON-Admin user sees when tries to access DC server



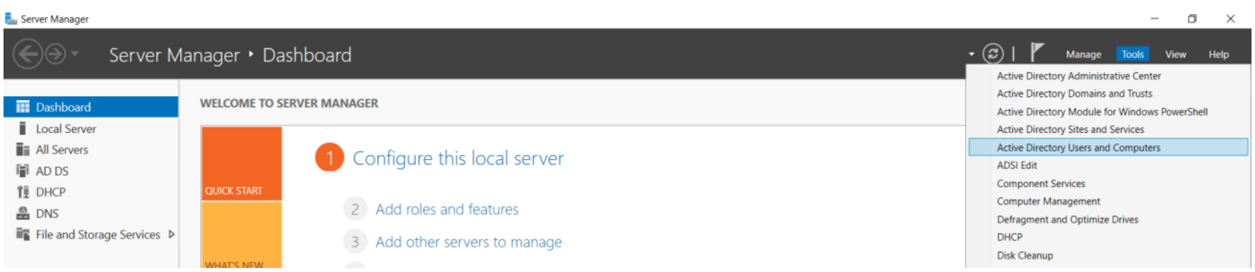
Enter your Credentials (Admin) – Admin_username: xxx & Admin_Password: xxx



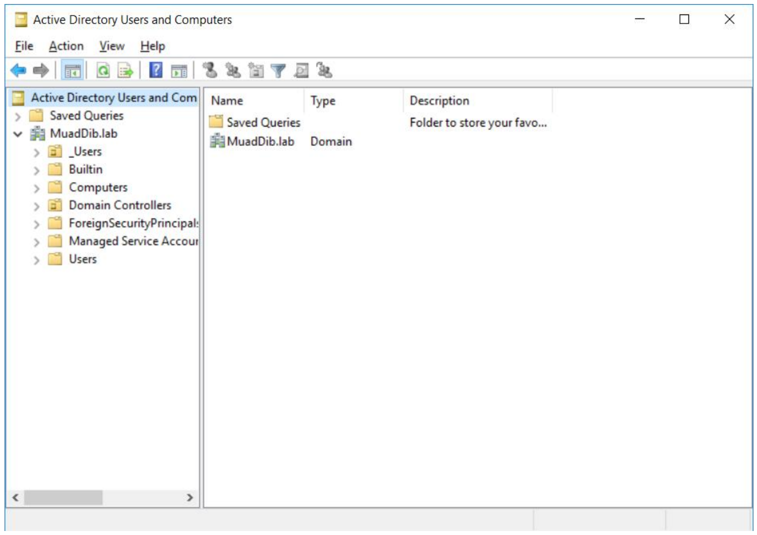
Click YES



Server Manager Dashboard – Click on Tools

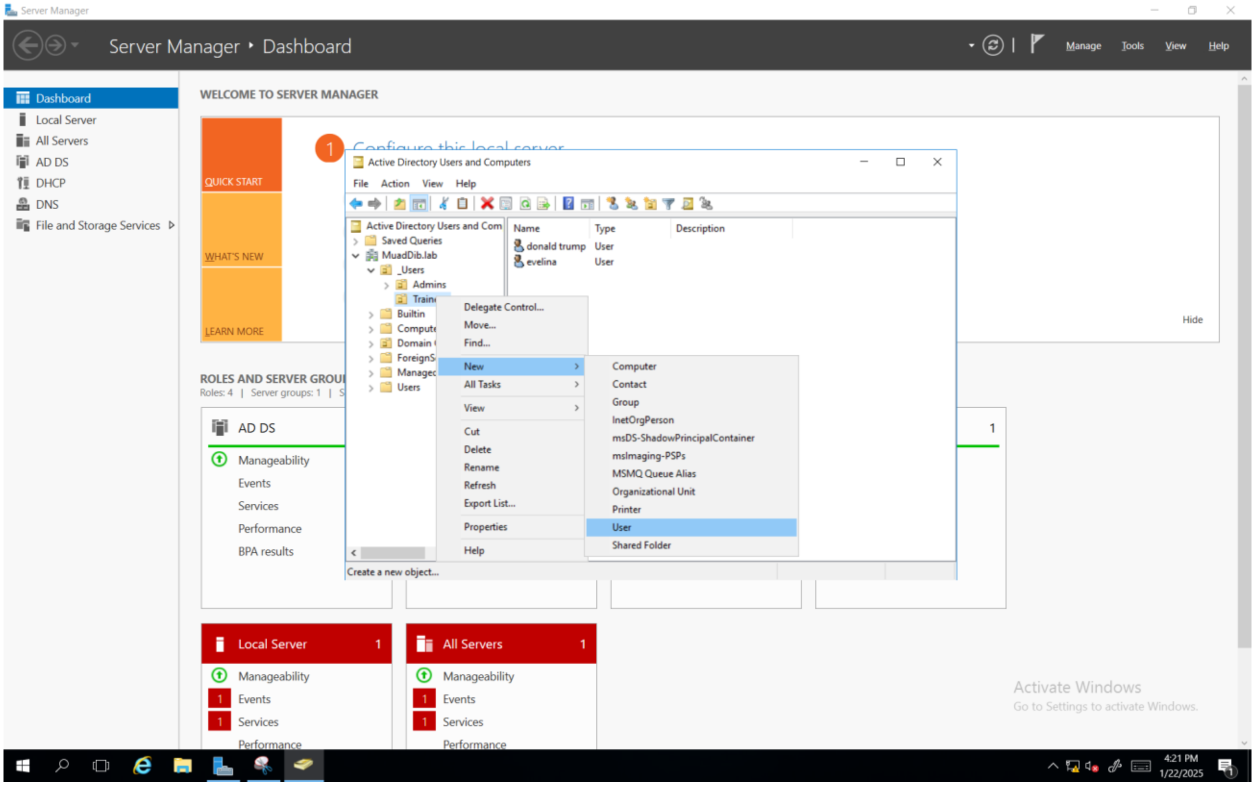


Choose Active Directory Users and Computers

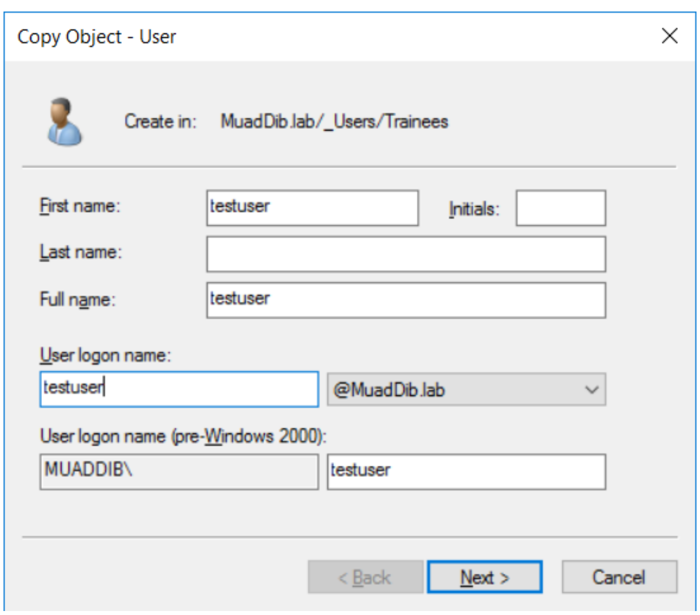


This is the Window that it opens

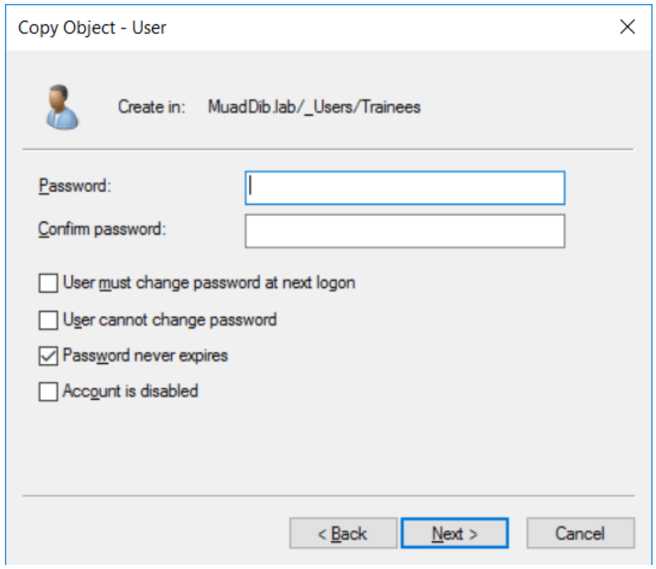
Create a New User:



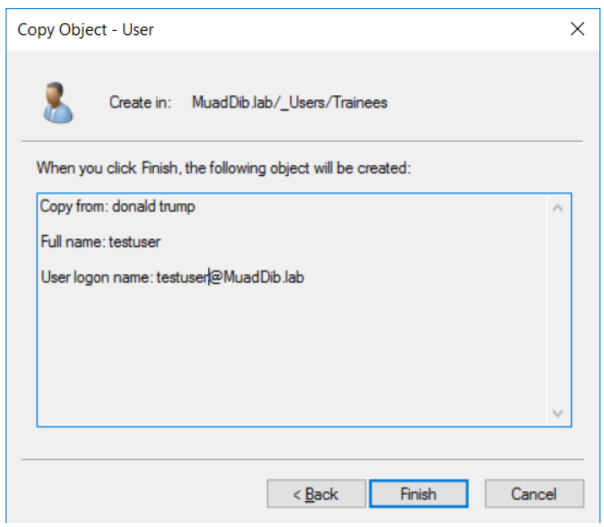
MuadDib.lab > _Users > Trainees: Right Click: New > User



First name, Last name, User logon name etc > Next



Password and if we need check the corresponding boxes



And “Finish”

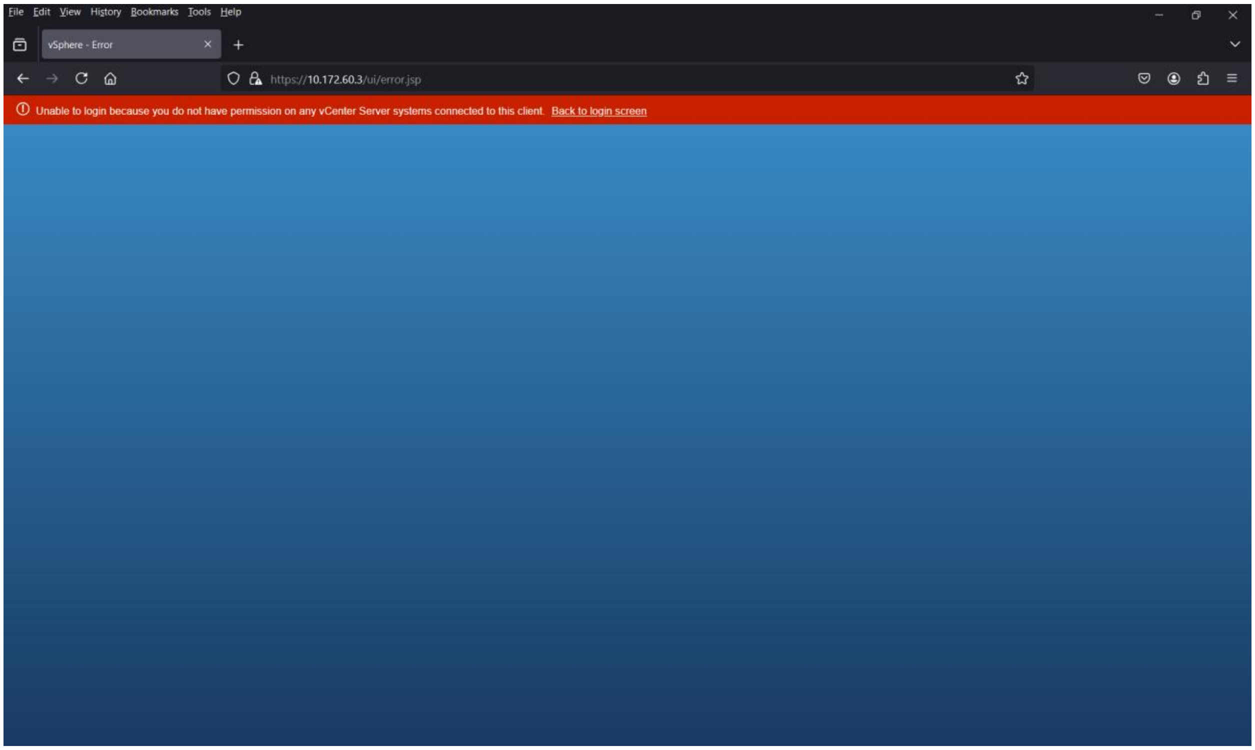
(In this situation in order to create the new user we copied the user donaldtrump. This way we can save administration workload and create users more efficiently)

*Right now all the created users are able to access vCenter (**10.172.60.3**)*.

Access vCenter

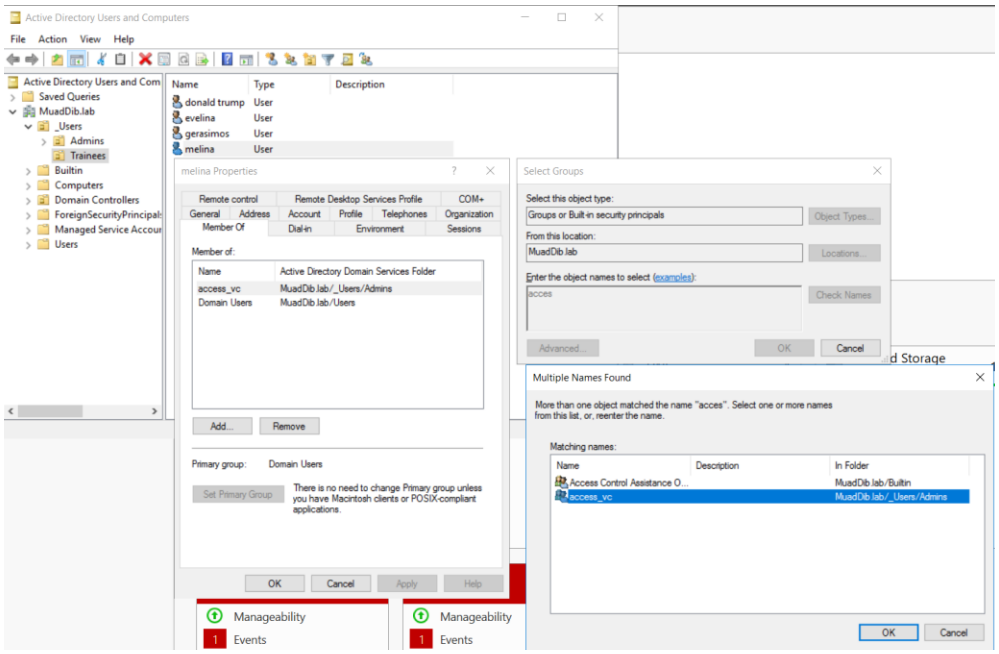
Error Message:

“Unable to login because you don’t have permission on any vCenter Server systems”



Solution:

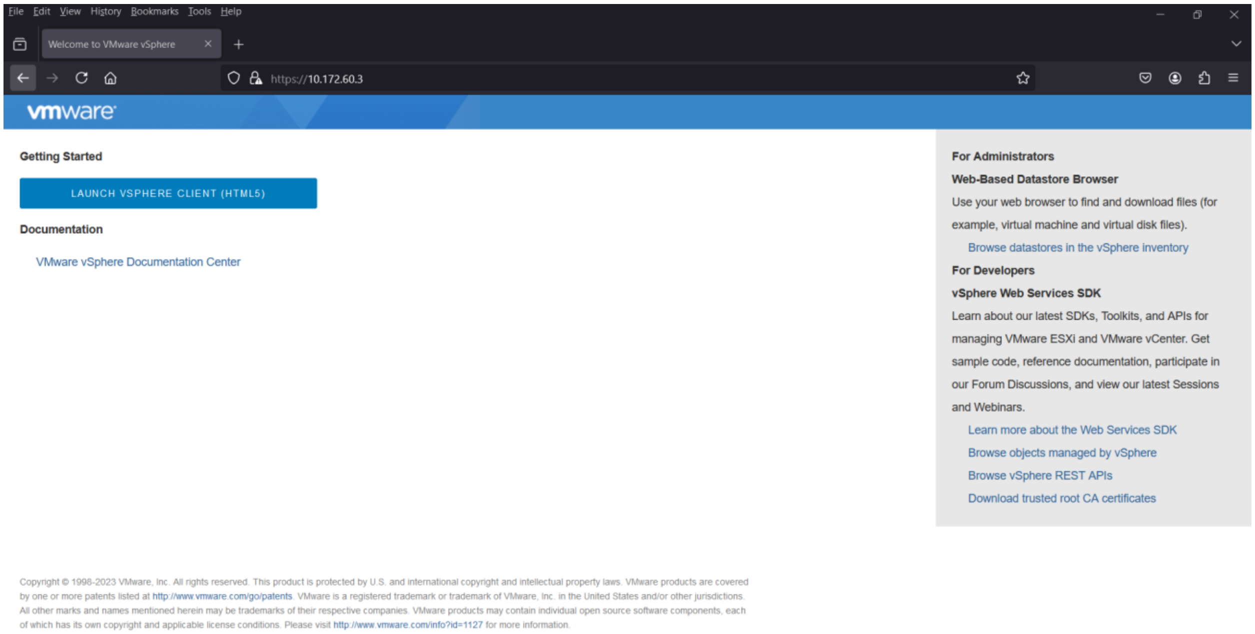
So we have to do the following steps on the Domain Controller:



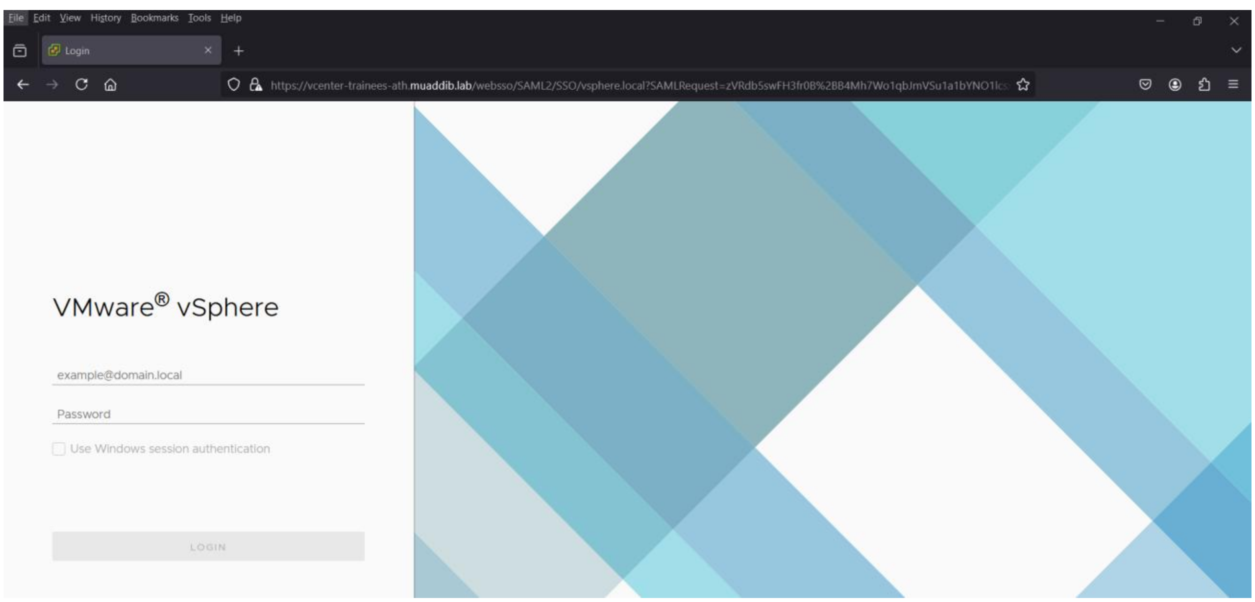
Choose the correct User: Double Click > Tab: Member Of > add: "Enter the object names" and click on Check Names > Choose the correct one (access_vc and <u>always Check Names</u>) and OK

The Following Configuration is only for DTS Training Lab Environment.

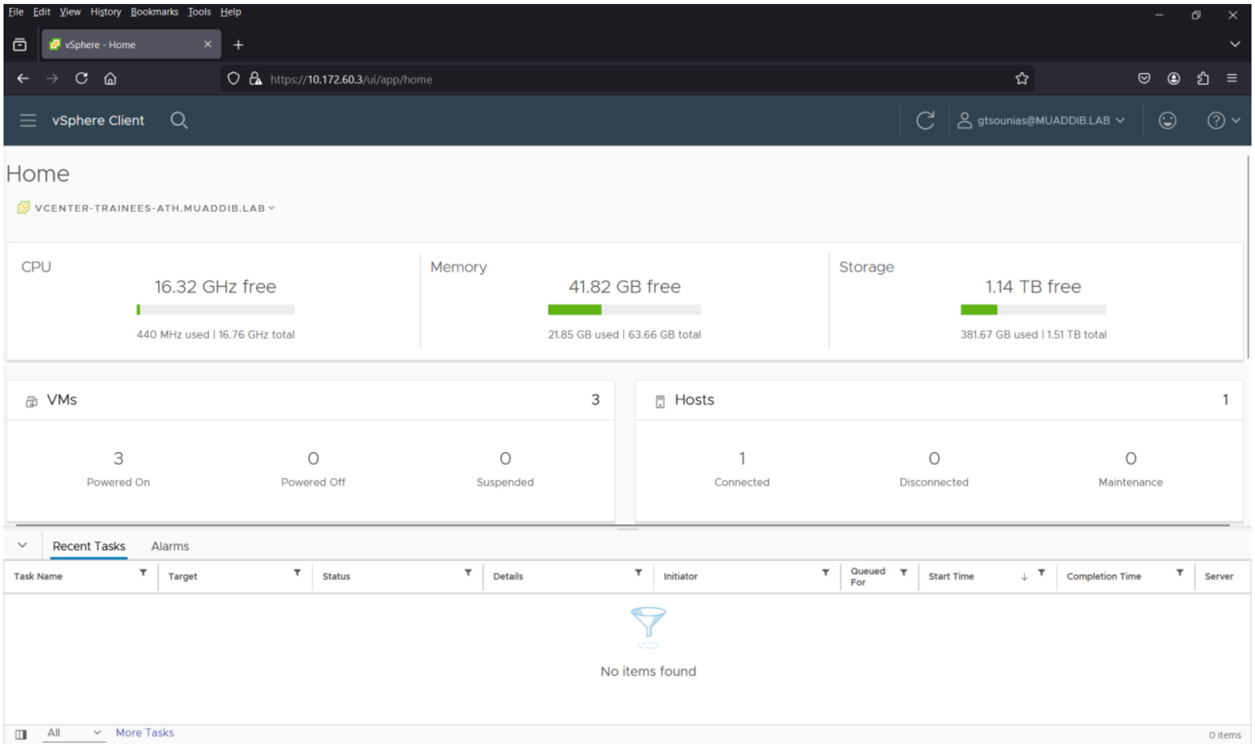
IP Address: 10.172.60.3



Web Browser: 10.172.60.3 > Press on “Launch vSphere Client”

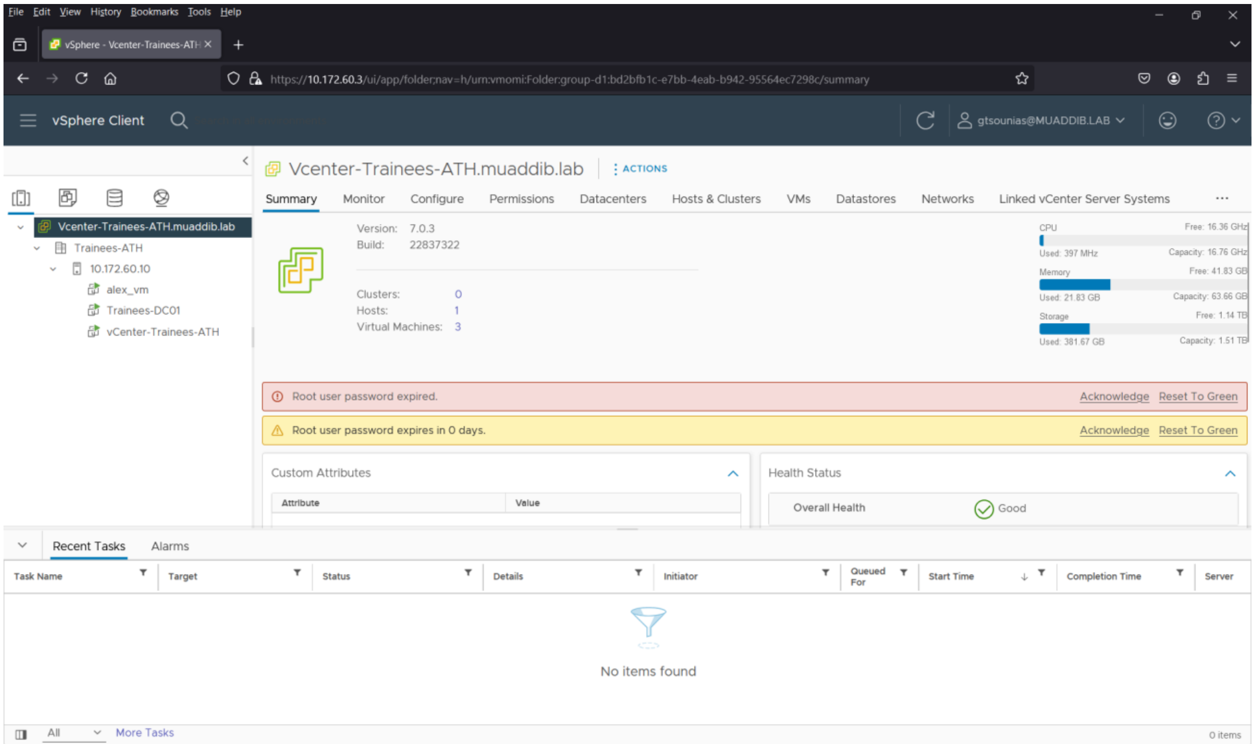


Enter Username: muaddib\“*your username*” & Password: “*your password*”



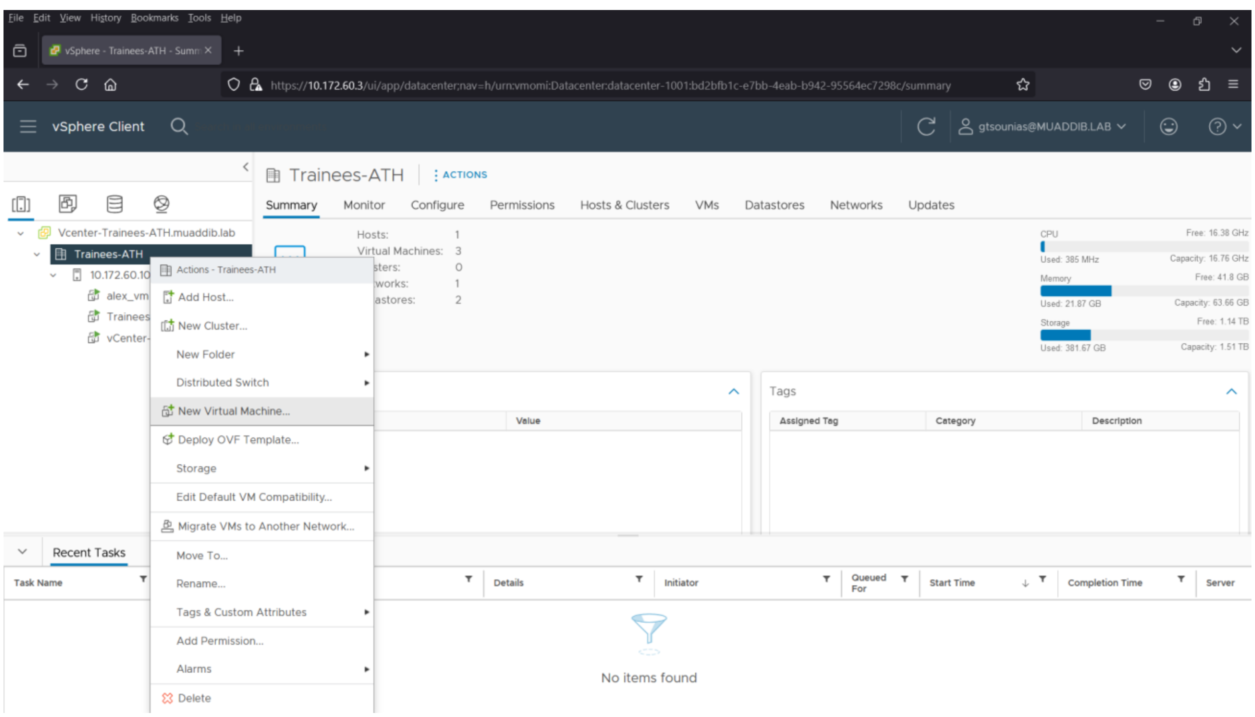
vCenter Home Screen

Here we can see general information about our vCenter resources like, CPU, Memory, and Storage usage. Moreover, we can find info about our VMs, Hosts, Recent Tasks and Alarms.

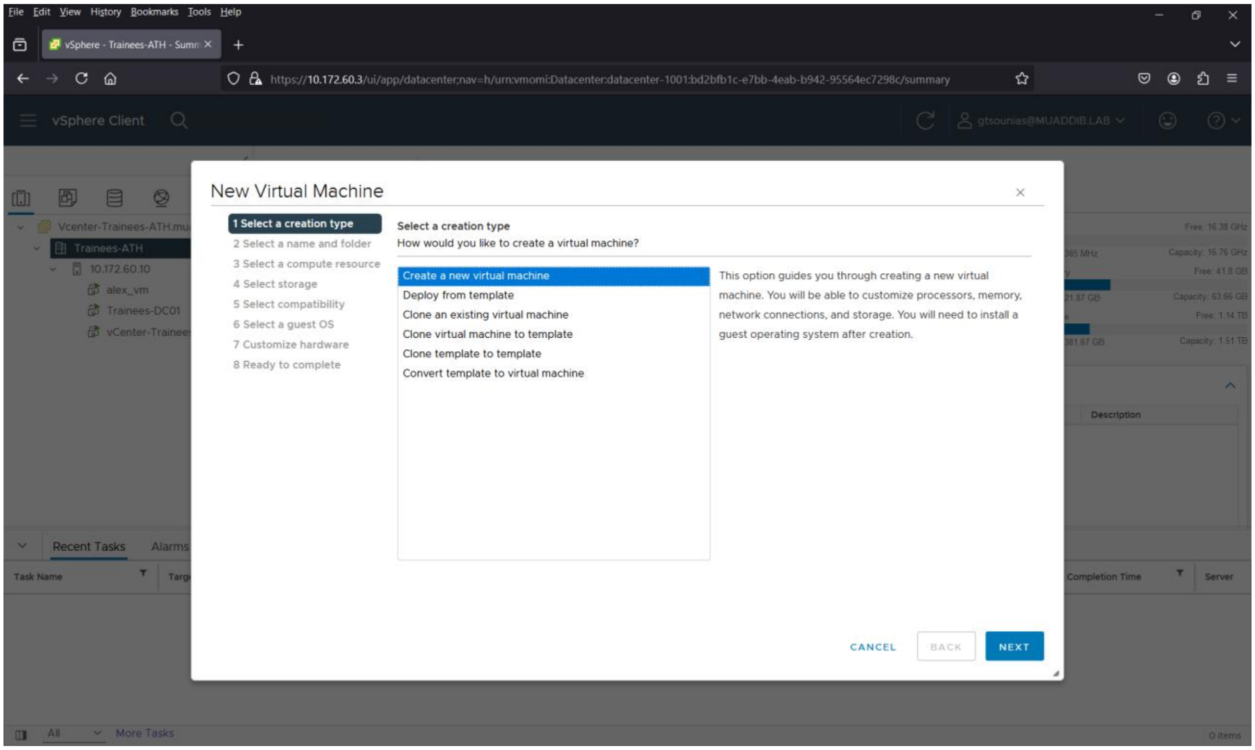


vCenter Main Page

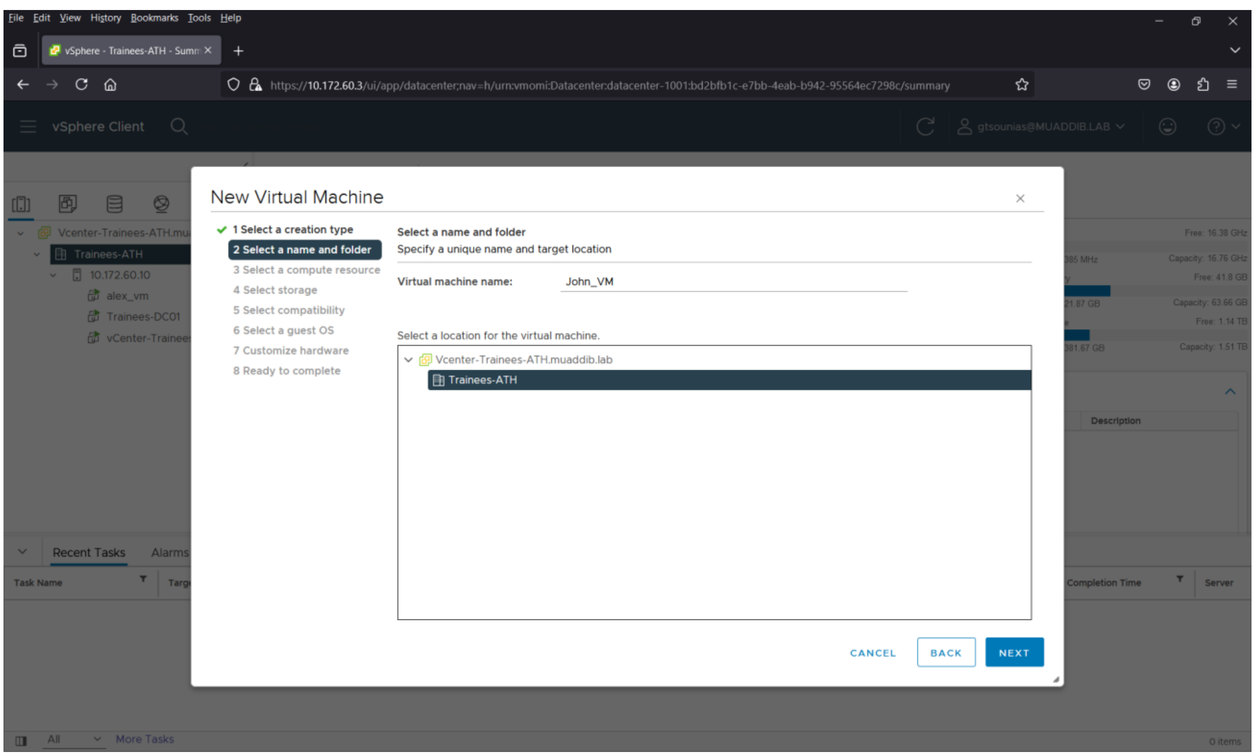
How to Create a New Virtual Machine



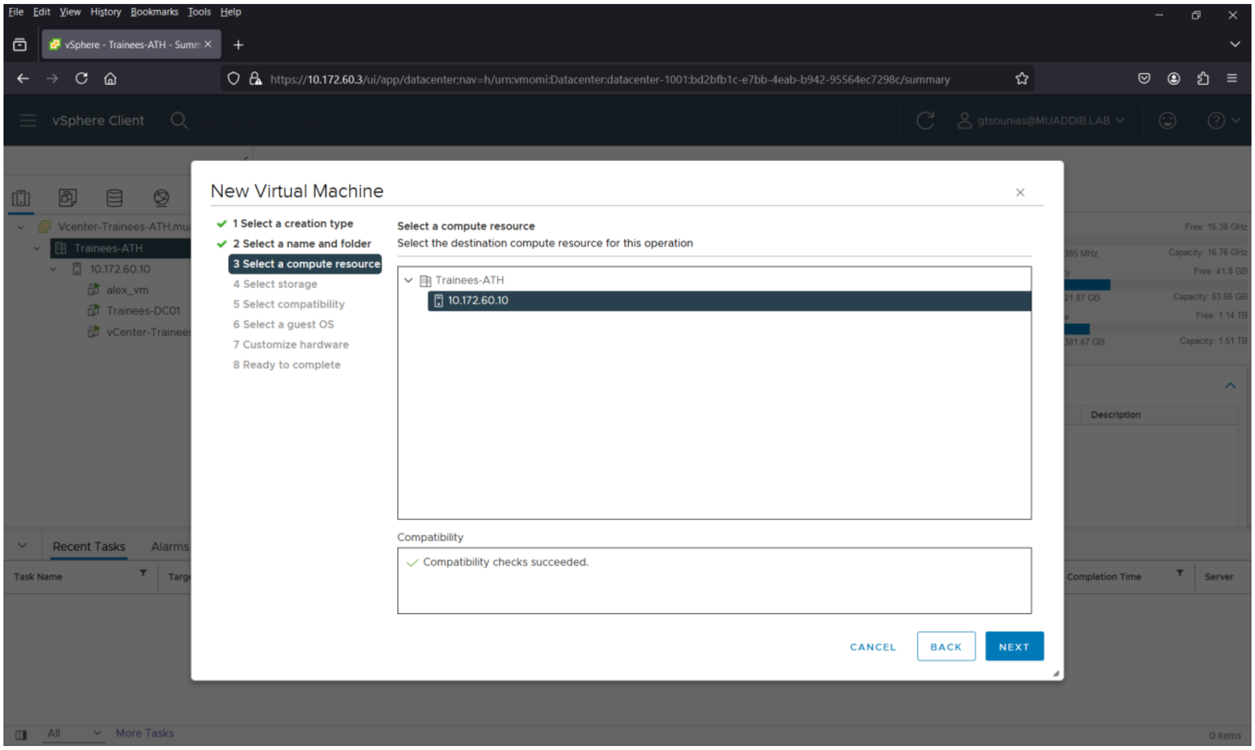
Right click on the Domain > New Virtual Machine

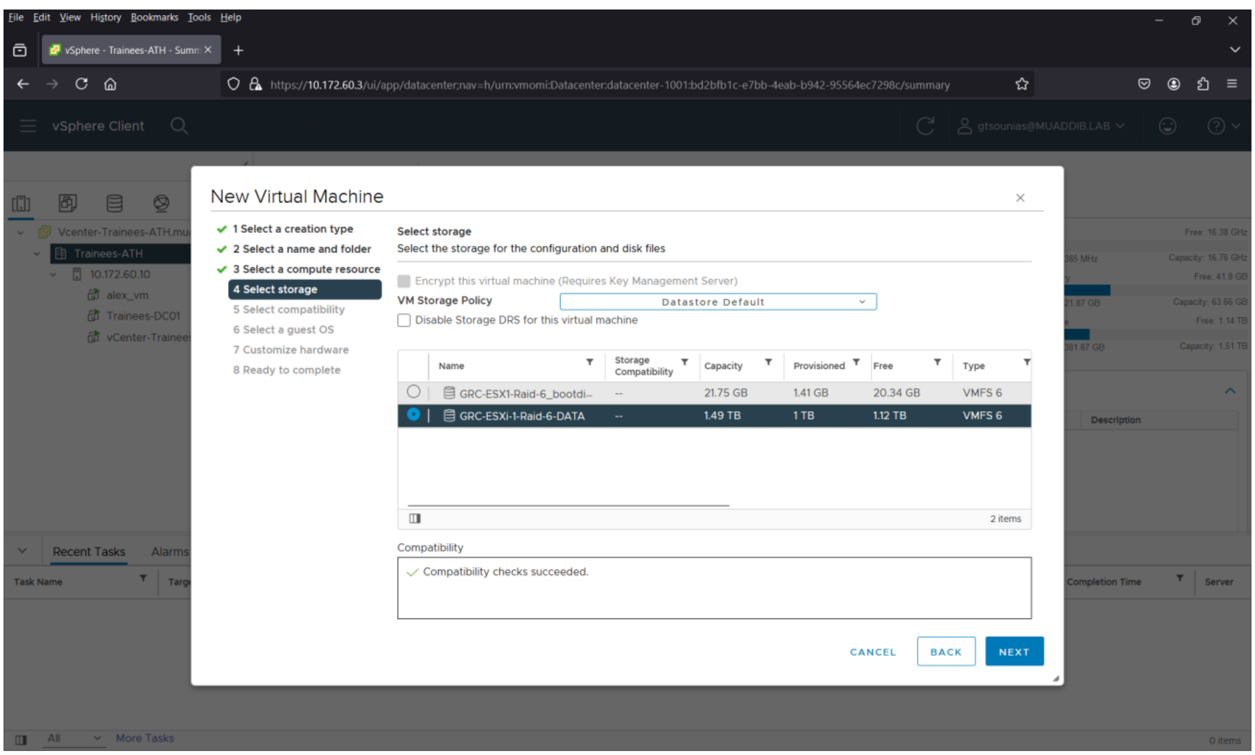


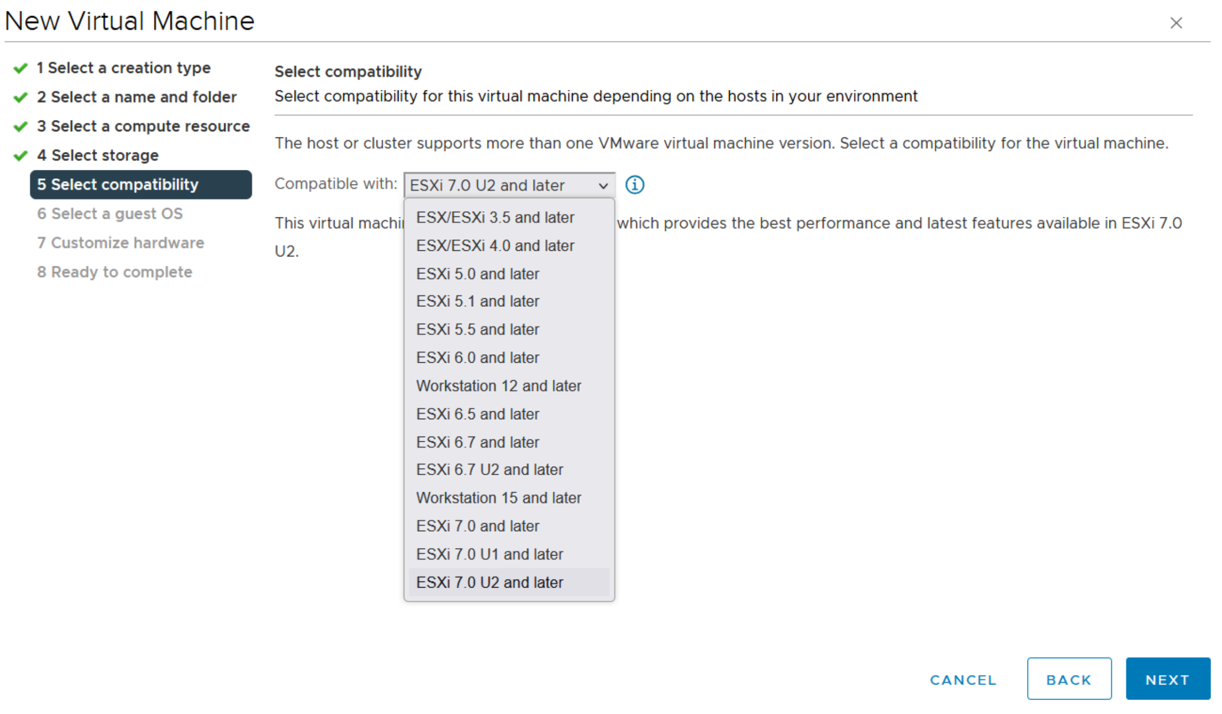
Choose “Create a New Virtual Machine”



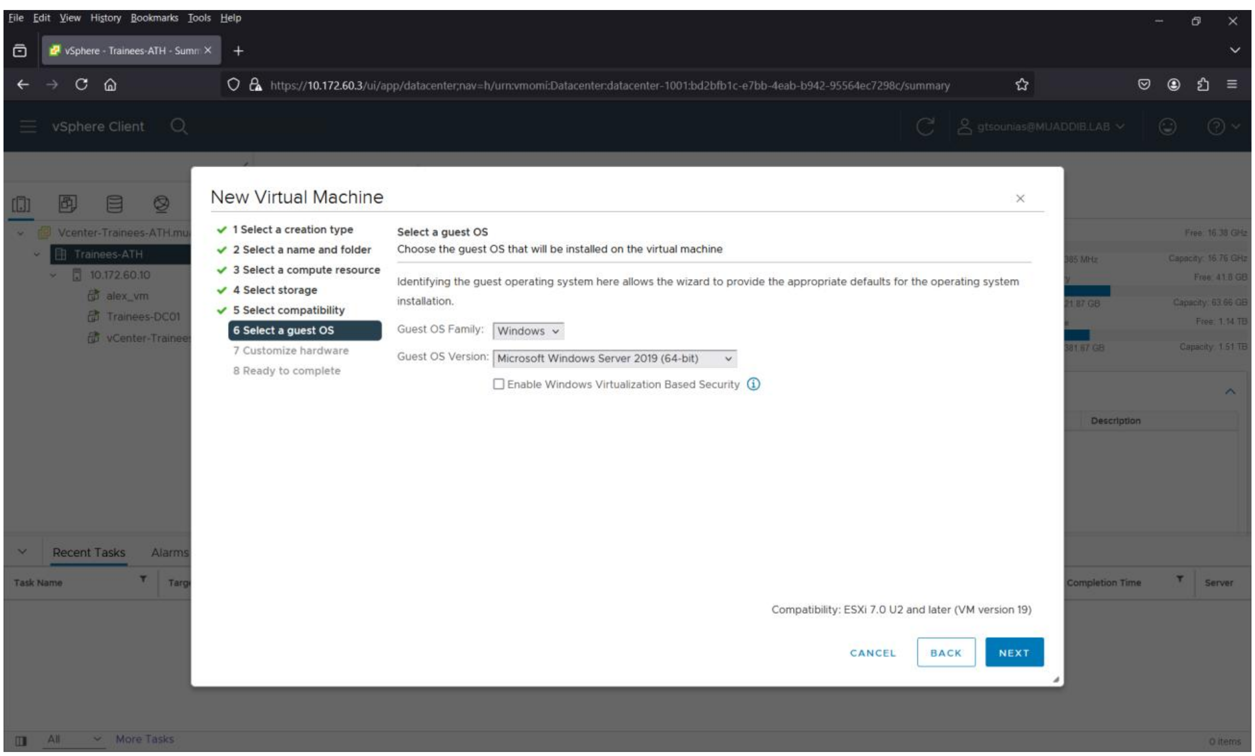
Give a name to the VM & choose the location


New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select a compute resource
Select the destination compute resource for this operation
Trainees-ATH
10.172.60.10
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT

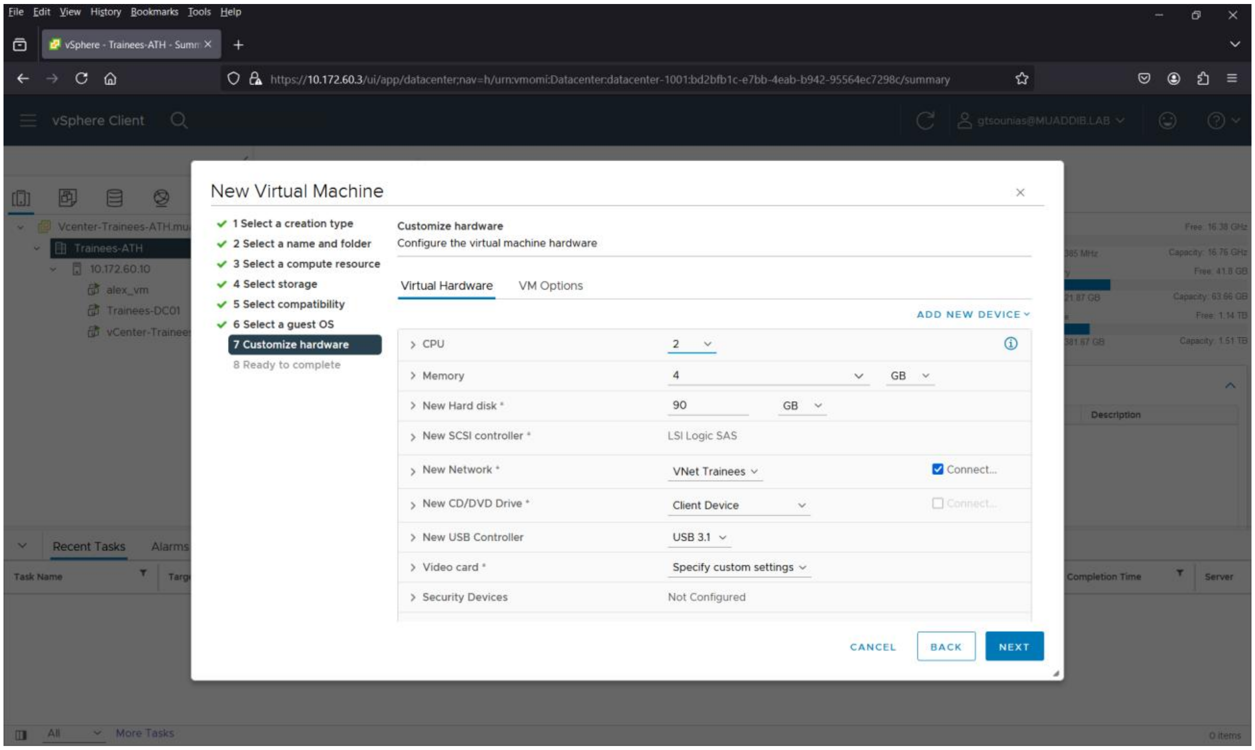

New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select storage
Select the storage for the configuration and disk files
Encrypt this virtual machine (Requires Key Management Server)
VM Storage Policy
Datastore Default
Disable Storage DRS for this virtual machine
Name
Storage Compatibility
Capacity
Provisioned
Free
Type
GRC-ESX1-Raid-6_bootdi...
21.75 GB
1.41 GB
20.34 GB
VMFS 6
GRC-ESXi-1-Raid-6-DATA
1.49 TB
1 TB
1.12 TB
VMFS 6
2 items
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT



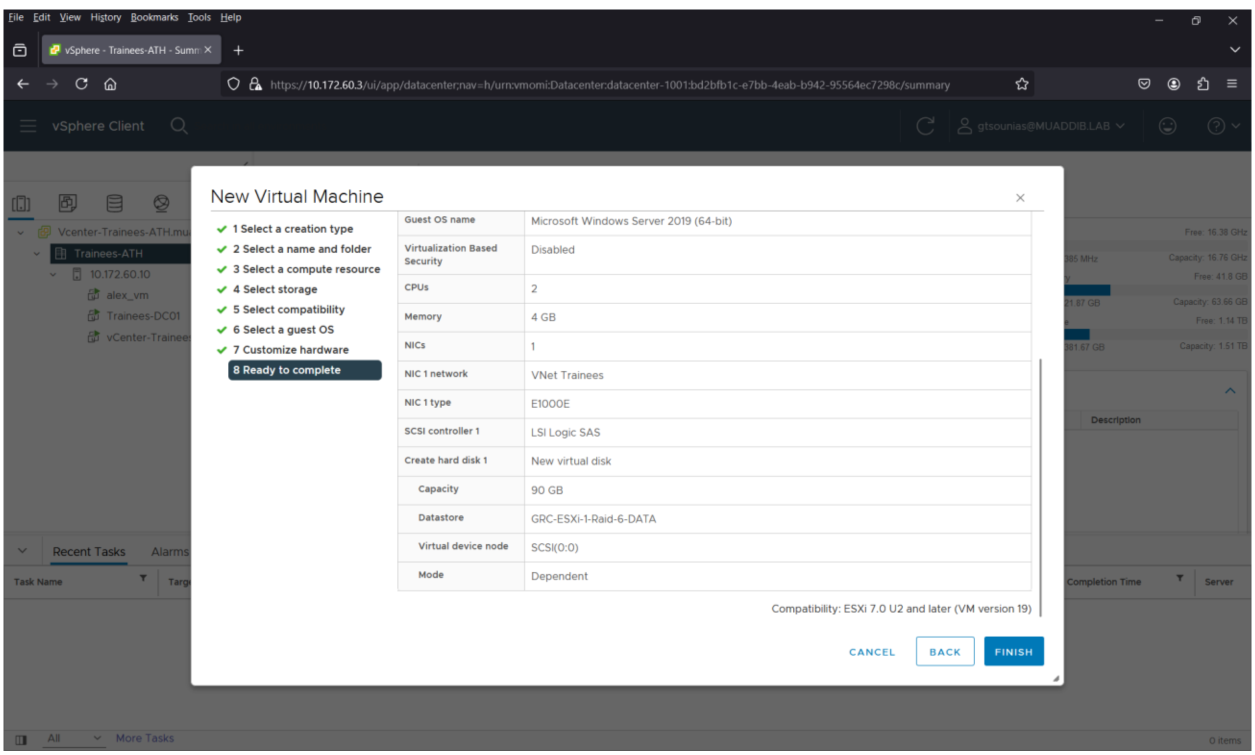
From the list choose: ESXi 7.0 U2 and later



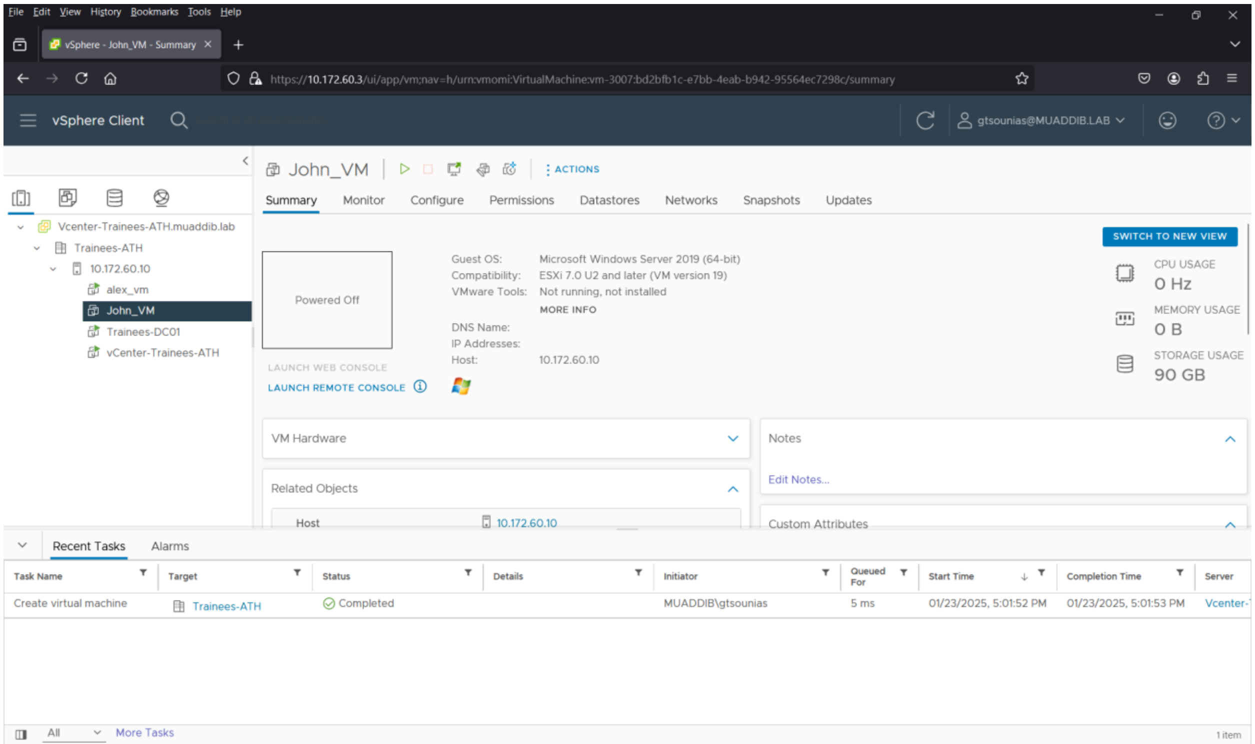
From the List choose Operating System and Version



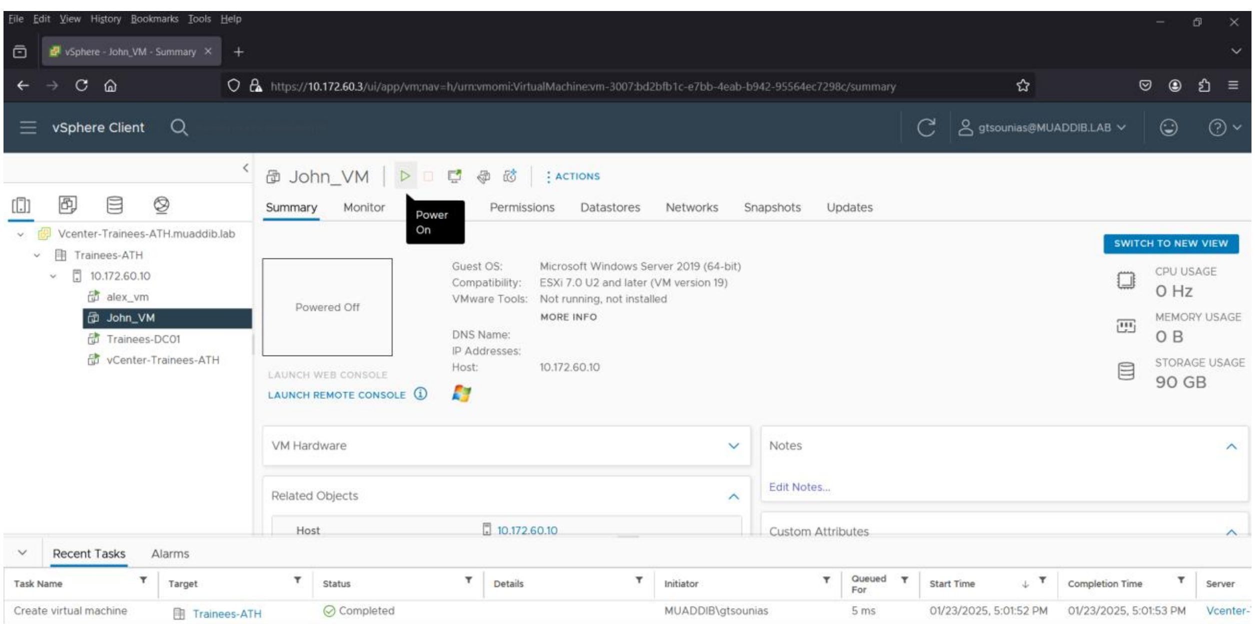
Customize the Hardware



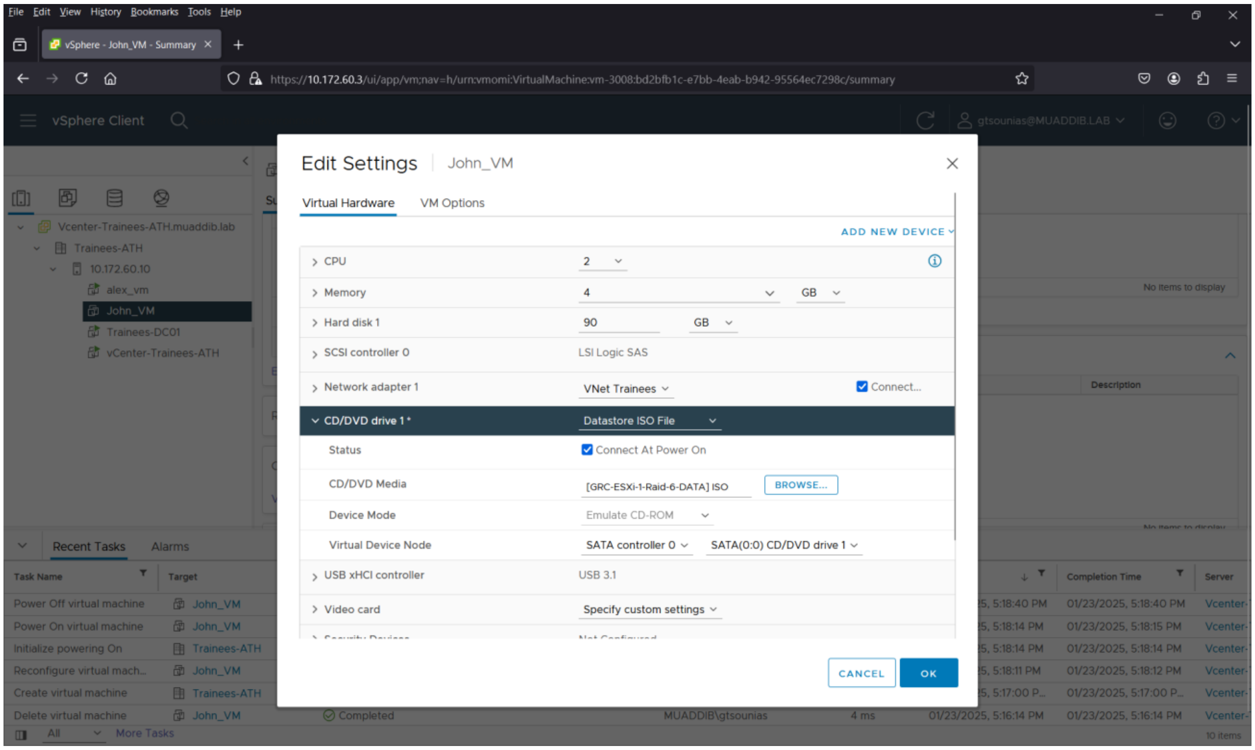
Final Check & Finish



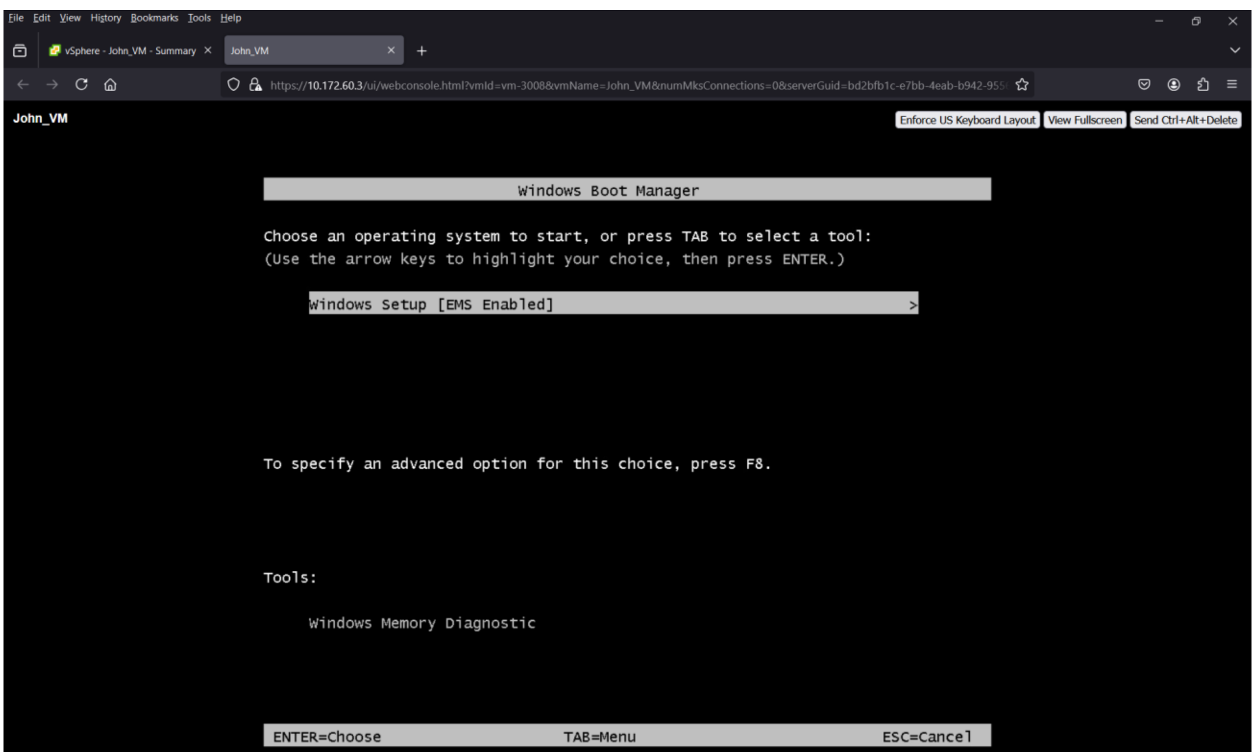
We can see the newly created VM and in Recent task the VM creation



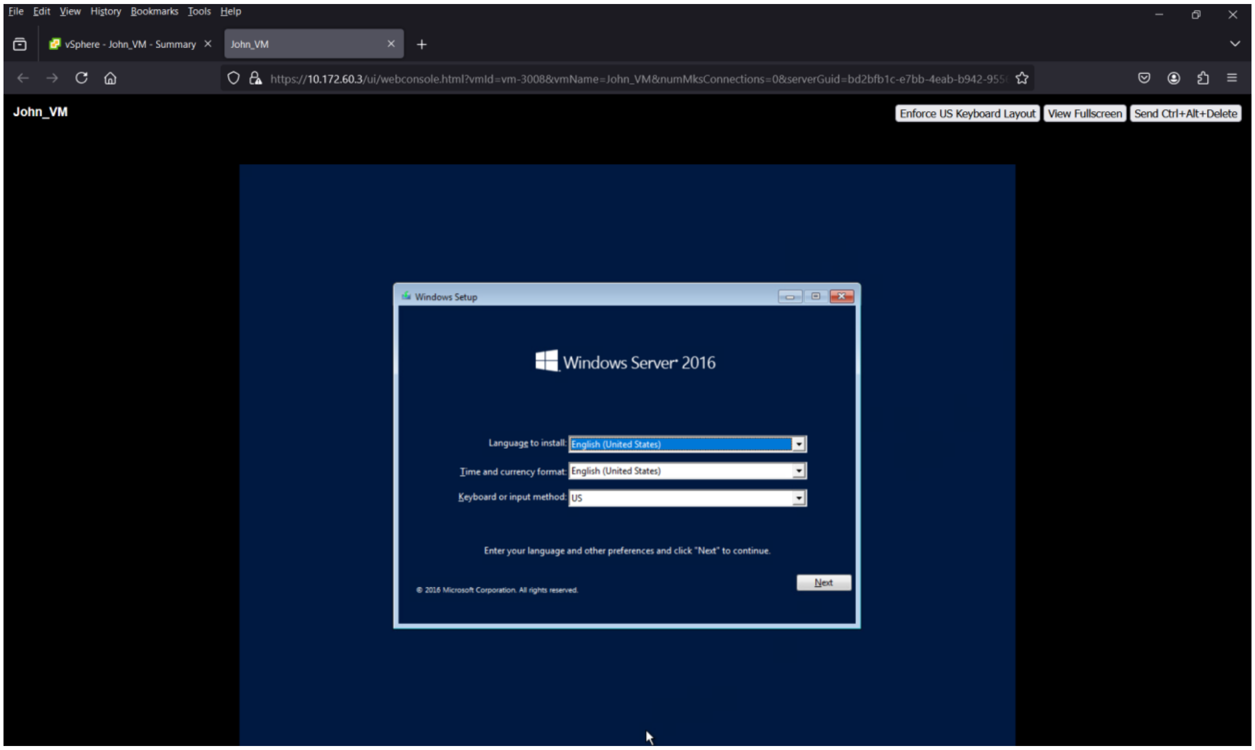
Power On the VM



Very important step is to mount the correct ISO file: CD/DVD > Datastore ISO File > Connect



Press Enter



Begin the Windows Server OS Installation Process